



Name : Yeamin Rahman Mahid
ID: 2023-3-60-375

Submitted to:
Md. Adnan Morshed
Lecturer,
Department of Computer Science
And Engineering

1. Brighten the following image

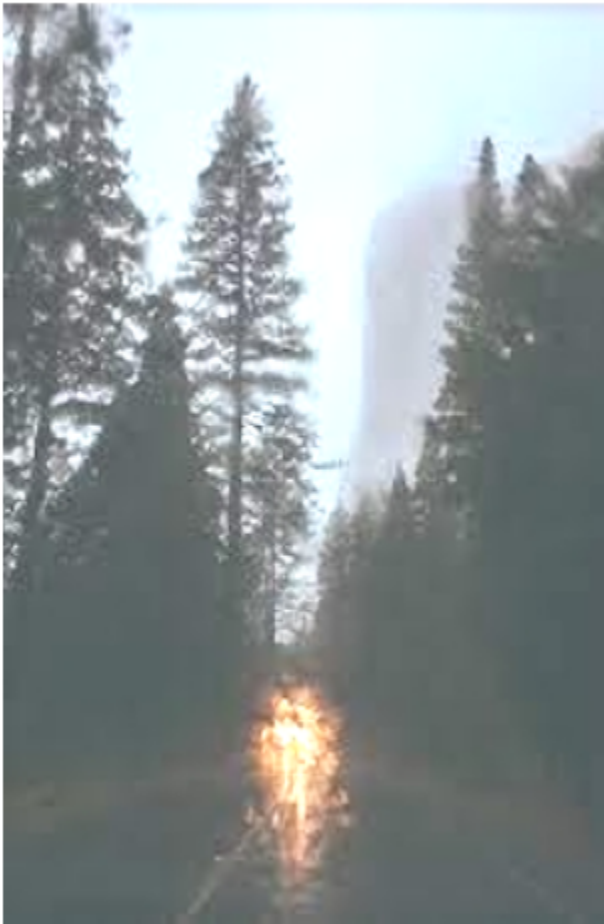
```
img_bright = cv2.imread("brighten.jpeg")
img_bright = cv2.cvtColor(img_bright, cv2.COLOR_BGR2RGB)

show_image(img_bright, "Original Image")
brightness_value = 70

brightened_img = cv2.add(img_bright, np.ones(img_bright.shape,
dtype=np.uint8) * brightness_value)

show_image(brightened_img, "Brightened Image")
```

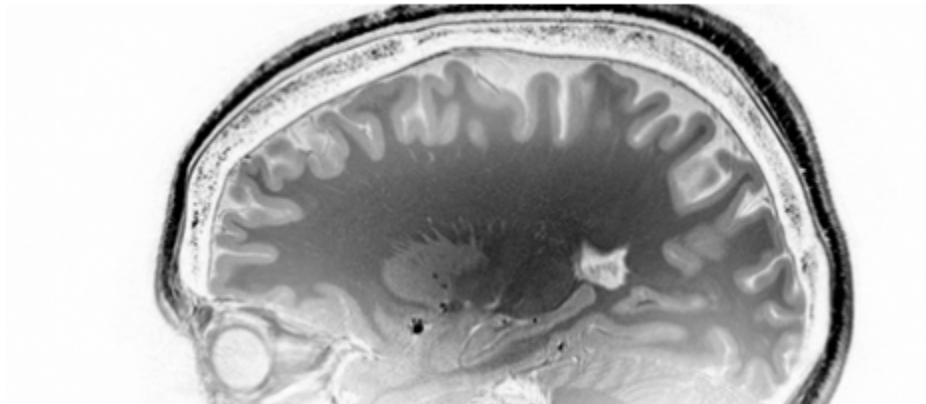
Brightened Image



2. Find the digital negative of the image.

```
negative_img = 255 - img_negative  
  
show_image(negative_img, "Digital Negative Image", cmap="gray")
```

Digital Negative Image



3. Adjust the contrast of the following image.

```
img = cv2.imread("contrast.jpeg", cv2.IMREAD_GRAYSCALE)

equalized = cv2.equalizeHist(img)

plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(img, cmap='gray')
plt.title("Original")
plt.axis('off')

plt.subplot(1,2,2)
plt.imshow(equalized, cmap='gray')
plt.title("Contrast Adjusted")
plt.axis('off')

plt.show()
```

Original



Contrast Adjusted



4. Use contrast stretching on the image.

```
min_pixel = np.min(img_stretch)
max_pixel = np.max(img_stretch)

contrast_stretched = ((img_stretch - min_pixel) / (max_pixel - min_pixel))
* 255
contrast_stretched = contrast_stretched.astype(np.uint8)

show_image(contrast_stretched, "Contrast Stretched Image", cmap="gray")
```

Contrast Stretched Image



5. Apply bit plane slicing on the image.

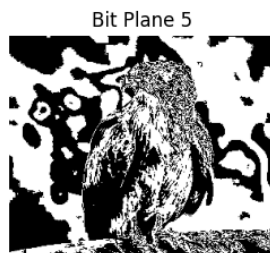
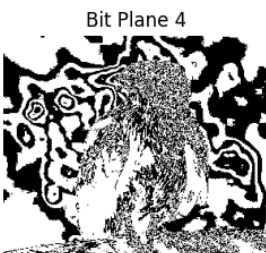
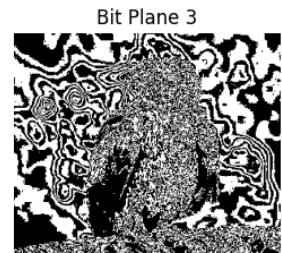
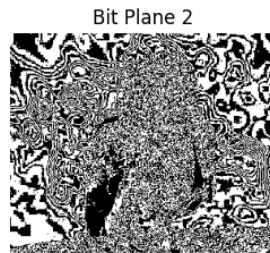
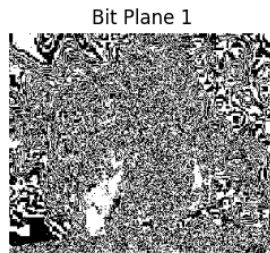
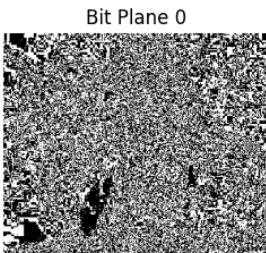
```
bit_planes = []

for i in range(8):
    bit_plane = cv2.bitwise_and(img_bitplane, 2**i)
    bit_plane = np.where(bit_plane > 0, 255, 0).astype(np.uint8)
    bit_planes.append(bit_plane)

plt.figure(figsize=(14,8))

for i in range(8):
    plt.subplot(2,4,i+1)
    plt.imshow(bit_planes[i], cmap="gray")
    plt.title(f"Bit Plane {i}")
    plt.axis("off")

plt.show()
```



```
plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
plt.imshow(bit_planes[5], cmap="gray")
plt.title("Bit Plane 5")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(bit_planes[6], cmap="gray")
plt.title("Bit Plane 6")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(bit_planes[7], cmap="gray")
plt.title("Bit Plane 7")
plt.axis("off")

plt.show()
```

